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GoSMS - SMS Credit Reload Feature

TEST PLAN AND TEST CASES

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Contents

TESTING OVERVIEW....2
STATE TRANSITION ANALYSIS....2
Initial Loading And Brand Selection....2
SMS Credit Entry And Validation....3
Quotation Generation And Credit Allocation....3
Credit Balance Refresh And Email Delivery....4
Quotation History And Persistence....5
Test Cases....6
Initial Loading And Brand Selection....6
SMS Credit Entry and Validation....8
Quotation Generation and Credit Allocation....10
Credit Allocation Safeguards....12
Credit Balance Refresh and Email Delivery....13
Quotation History and Persistence....15
Error and Recovery Testing....18
Validation and Request Failures....18
Data Integrity....18
Independent Operations and Recovery....18
Transition Coverage Matrix....19

TESTING OVERVIEW

This test plan focuses on the SMS Credit Reload feature of GoSMS. Testing covers the process of selecting a client brand, entering SMS credit amounts for its outlets, generating a quotation, allocating credits, sending the quotation by email, and accessing quotation history.

Functional testing will verify that each part of the feature behaves as expected. Input validation and boundary testing will cover empty values, invalid entries, and credit amounts that exceed supported limits. Integration testing will examine how the frontend, API, Redis database, and email service interact during quotation processing and credit allocation.

Persistence testing will check whether credit balances and quotation history remain available after refreshing the application. Error and recovery testing will examine failed requests, unsuccessful email delivery, and retry behaviour, particularly whether repeated quotation requests can cause duplicate credit allocations.

State transition analysis will support these testing methods by identifying the different states the feature can enter and the actions that move it between them. The test cases will cover these transitions while avoiding unnecessary repetition.

STATE TRANSITION ANALYSIS

The SMS Credit Reload feature moves through several states as users select a brand, enter credit amounts, generate quotations, and access saved records. The analysis identifies the actions that trigger these transitions, their expected outcomes, and how the feature responds when an operation fails.

Initial Loading And Brand Selection

ID	Current State	Action or Condition	Expected Result
ST01	Loading	Brand and credit balance data load successfully	The brand selection screen is displayed.
ST02	Loading	Initial data loading fails	An error message appears with a Retry option.
ST03	Loading error	User selects Retry	Data loading is attempted again. The feature proceeds to brand selection if successful or displays the error again if unsuccessful.
ST04	No brand selected	User selects a brand	The selected brand is stored and Continue becomes enabled.
ST05	Brand selected	User selects Continue	The SMS Top Up screen opens, displaying the brand's outlets and existing credit balances.
ST06	SMS Top Up	User selects Change Brand	The feature returns to brand selection while retaining the current session's credit entries.
ST07	Brand selection	User returns to a previously selected brand	Previously entered credit amounts for that brand are restored within the same session.

SMS Credit Entry And Validation

ID	Current State	Action or Condition	Expected Result
ST08	SMS Top Up	User enters valid credit amounts for one or more outlets	The entered amounts are stored in the draft and the total is updated.
ST09	SMS Top Up	User leaves an amount empty or enters only spaces	The amount is treated as zero without triggering a validation error.
ST10	SMS Top Up	User enters a negative, decimal, or nonnumeric value	An inline validation error is displayed and quotation generation is prevented.
ST11	SMS Top Up	User enters an amount exceeding the safe integer limit	The amount is rejected with an error indicating that it is too large.
ST12	Invalid credit entry	User corrects the invalid amount	The validation error clears and the total is recalculated.
ST13	SMS Top Up	User attempts to generate a quotation when the total is zero	Generation is prevented and the user is prompted to allocate at least one SMS credit.
ST14	SMS Top Up	Selected brand has no configured outlets	Generation is prevented and an appropriate error is displayed.
ST15	Valid credit draft	User selects Generate Quotation	The quotation request is submitted and editing is disabled while processing.

Quotation Generation And Credit Allocation

ID	Current State	Action or Condition	Expected Result
ST16	Processing quotation	The server successfully creates the quotation and allocates credits	The quotation is displayed, credit balances are updated in Redis, and the quotation is saved in history.
ST17	Processing quotation	The server rejects the request with a recoverable client error	An error is displayed, and the user can correct the draft before trying again.
ST18	Processing quotation	A network failure, server error, or conflicting operation response occurs	The allocation status is treated as uncertain. The draft remains locked and Retry Quotation becomes available.
ST19	Quotation awaiting retry	User retries an operation that was already allocated successfully	The server returns the original quotation without allocating the credits again.
ST20	Quotation awaiting retry	The retry also fails without confirming the result	The feature remains in the retry state, preserving the original operation details.

ST21	Previously recorded operation	The same operation ID is submitted with different credit allocation details	The server rejects the conflicting request without changing the existing allocation.
ST22	Processing quotation	Quotation history storage is unavailable or has an invalid structure	The request fails without partially updating credit balances or saving a quotation.
ST23	Processing quotation	The requested allocation would cause a stored balance to exceed its supported limit	The request is rejected before any credits are allocated.

Credit Balance Refresh And Email Delivery

ID	Current State	Action or Condition	Expected Result
ST24	Quotation generated	The feature starts refreshing credit balances and sending the quotation email	Both operations proceed independently without affecting the confirmed credit allocation.
ST25	Refreshing credit balances	Updated balances are retrieved successfully	The latest credit balances are displayed.
ST26	Refreshing credit balances	The balance request fails	An error message is displayed, and unavailable balances are shown instead of outdated values.
ST27	Sending quotation email	The email is delivered successfully	The email status changes to Sent.
ST28	Sending quotation email	Email delivery fails	The quotation remains valid, allocated credits are retained, and Retry Email becomes available.
ST29	Email delivery failed	User selects Retry Email	Another delivery attempt begins without generating a new quotation or allocating additional credits.
ST30	Retrying email delivery	The retry succeeds	The email status changes to Sent.
ST31	Quotation displayed	User selects Back to SMS Top Up	The credit entry screen opens with a fresh draft and the previous allocation is retained in the stored balances.

Quotation History And Persistence

ID	Current State	Action or Condition	Expected Result
ST32	SMS Top Up	User selects History	The quotation history screen opens and begins loading saved quotations for the selected brand.
ST33	Loading history	Saved quotations are retrieved successfully	The history list displays the available quotations.
ST34	Loading history	No quotations exist for the selected brand	An empty history message is displayed.
ST35	Loading history	The history request fails and no local quotations are available	An error message appears with a Retry History option.
ST36	Loading history	The history request fails but quotations from the current session are available	Locally available quotations are displayed with a notice that the full history could not be retrieved.
ST37	History loading error	User selects Retry History	The application attempts to retrieve the saved quotations again.
ST38	Quotation history	User selects a saved quotation	The selected quotation opens in read-only mode without changing credit balances.
ST39	Viewing saved quotation	User selects Back to History	The quotation history screen is displayed again.
ST40	Quotation history	User selects Back to SMS Top Up	The feature returns to the credit entry screen.
ST41	Previously saved quotations and balances	User refreshes or reopens the application	Credit balances and saved quotation history remain available from persistent storage, while unfinished draft entries are not restored after a page reload.

Test Cases

The test cases examine how the SMS Credit Reload feature behaves under normal and unexpected conditions. Each case includes the starting conditions, actions performed, and expected results, with references to the relevant state transitions.

Initial Loading And Brand Selection

TC01 – Successful Initial Loading

Precondition: The business and credit balance services are available and contain valid data.

Steps:

1. Open the SMS Credit Reload feature.
2. Allow the initial data requests to complete.
3. Check the brand selection screen.

Expected Result: The available brands are displayed, the initial credit balance data is loaded, and no loading errors appear.

Transitions Covered: ST01

TC02 – Initial Loading Failure and Recovery

Precondition: The feature is opened while either the business or credit balance service is unavailable.

Steps:

1. Open the feature with one of the required services unavailable.
2. Observe the error displayed.
3. Restore the unavailable service.
4. Select Retry.

Expected Result: The feature displays an error screen with a Retry option when loading fails. Once the service is restored, retrying loads the required data and displays the brand selection screen.

Transitions Covered: ST02, ST03

TC03 – Brand Selection and Navigation

Precondition: The brand selection screen has loaded successfully.

Steps:

1. Check the Continue button before selecting a brand.
2. Select an available brand.
3. Select Continue.
4. Inspect the outlets and credit balances displayed.

Expected Result: Continue remains disabled until a brand is selected. After continuing, the SMS Top Up screen displays only the selected brand's outlets, with their corresponding existing credit balances.

Transitions Covered: ST04, ST05

TC04 – Brand Switching and Draft Retention

Precondition: At least two brands are available, each with configured outlets.

Steps:

1. Select Brand A and enter credit amounts for its outlets.
2. Select Change Brand and open Brand B.
3. Enter different credit amounts for Brand B.
4. Switch back to Brand A and inspect its entries.
5. Return to Brand B and inspect its entries.

Expected Result: Each brand retains its own credit entries while the current application session remains active. Switching between brands restores the correct draft without mixing or overwriting amounts from another brand.

Transitions Covered: ST06, ST07

SMS Credit Entry and Validation

TC05 – Credit Entry and Total Calculation

Precondition: A brand with multiple outlets has been selected, and the SMS Top Up screen is displayed.

Steps:

1. Enter 100 SMS credits for the first outlet.
2. Enter 250 SMS credits for the second outlet.
3. Leave another outlet's credit field empty.
4. Enter only spaces in another available credit field.
5. Check the total displayed.
6. Change the first outlet's amount to 150 and check the total again.

Expected Result: The feature accepts valid whole numbers and treats empty or whitespace entries as zero. The initial total is 350 credits and updates to 400 after the change. No validation errors are displayed.

Transitions Covered: ST08, ST09

TC06 – Invalid Input and Correction

Precondition: The SMS Top Up screen is open with at least one configured outlet.

Steps:

1. Enter a negative credit amount, such as -10.
2. Replace it with a decimal value, such as 25.5.
3. Replace it with alphabetic characters, such as abc.
4. Attempt to generate a quotation while an invalid value is present.
5. Correct the entry to a valid whole number.

Expected Result: Each invalid entry produces an inline validation error. Quotation generation is prevented while invalid amounts remain. Once corrected, the error clears and the total reflects the valid amount.

Transitions Covered: ST10, ST12

TC07 – Credit Amount Boundary

Precondition: The SMS Top Up screen is open with an outlet available for credit entry.

Steps:

1. Enter 9007199254740992, which exceeds JavaScript's maximum safe integer.
2. Observe the validation message.
3. Attempt to generate a quotation.
4. Replace the amount with a valid value within the supported limit.

Expected Result: The oversized amount is rejected with an appropriate validation message, and quotation generation is prevented. Replacing it with a valid amount clears the error.

Transitions Covered: ST11

TC08 – Zero Credit Allocation

Precondition: A brand with configured outlets is selected.

Steps:

1. Leave all outlet credit fields empty.
2. Select Generate Quotation.
3. Enter zero in each credit field.
4. Select Generate Quotation again.

Expected Result: Both attempts are rejected because the total allocation is zero. The feature displays a message requesting at least one SMS credit, and no quotation or credit allocation is created.

Transitions Covered: ST13

TC09 – Brand Without Configured Outlets

Precondition: A brand exists in the test data without any configured outlets.

Steps:

1. Select the brand without configured outlets.
2. Open the SMS Top Up screen.
3. Attempt to generate a quotation.

Expected Result: No outlet entries are available, and quotation generation is prevented. The feature displays a message indicating that no outlets are configured for the selected brand. No quotation is generated.

Transitions Covered: ST14

Quotation Generation and Credit Allocation

TC10 – Successful Quotation Generation

Precondition: A brand with multiple outlets is selected, and valid credit amounts have been entered.

Steps:

1. Enter 100 SMS credits for the first outlet and 200 for the second.
2. Select Generate Quotation.
3. Check whether credit fields and navigation controls are disabled while the request is processing.
4. Examine the generated quotation and its allocation details.
5. Check the updated credit balances in Redis.
6. Verify that the quotation has been saved in history.

Expected Result: Editing is disabled during processing. The generated quotation contains the correct outlet allocations and a total of 300 SMS credits. The corresponding balances increase by the allocated amounts, and the quotation is saved in history.

Transitions Covered: ST15, ST16

TC11 – Server Rejection

Precondition: The quotation API is available, and a valid brand with configured outlets exists.

Steps:

1. Submit a quotation request through the API containing an outlet that does not belong to the selected brand.
2. Observe the server response.
3. Check whether any credit balances or quotation records have changed.
4. Correct the invalid allocation details and submit a valid request.

Expected Result: The server rejects the invalid request with an appropriate client error. No credits are allocated and no quotation is saved for the rejected request. The corrected request can be submitted without remaining locked in an uncertain processing state.

Transitions Covered: ST17

TC12 – Failed Request and Retry

Precondition: A valid credit allocation has been entered, and the quotation request can be interrupted to simulate a network or server failure.

Steps:

1. Select Generate Quotation.
2. Simulate a network failure or server error before the allocation result can be confirmed.
3. Check the error message and available actions.
4. Attempt to modify the credit amounts or switch brands.
5. Select Retry Quotation while the service remains unavailable.
6. Check the feature's state after the unsuccessful retry.

Expected Result: The feature retains the original operation ID and allocation details. Editing and brand switching remain unavailable while the outcome is uncertain. Retry Quotation remains available after another failed attempt, allowing the same operation to be retried without creating a new allocation request.

Transitions Covered: ST18, ST20

TC13 – Duplicate Credit Allocation Prevention

Precondition: A valid quotation request can be processed successfully while its response to the frontend is deliberately interrupted.

Steps:

1. Enter valid credit amounts and select Generate Quotation.
2. Allow the server to allocate the credits and save the quotation.
3. Simulate a lost response so the frontend cannot confirm the successful operation.
4. Select Retry Quotation.
5. Verify that the retry uses the original operation ID.
6. Compare the stored balances and quotation history before and after the retry.

Expected Result: The server recognises the repeated operation and returns the original quotation. Credits are allocated only once, and no duplicate quotation is created in history. The feature displays the confirmed quotation after the successful retry.

Transitions Covered: ST19

Credit Allocation Safeguards

TC14 – Conflicting Operation ID

Precondition: A quotation has been successfully generated and its operation ID is available.

Steps:

1. Submit a valid quotation request through the API and record its operation ID.
2. Confirm that the quotation was saved and the credits were allocated.
3. Submit another request using the same operation ID but different credit amounts.
4. Examine the server response.
5. Compare the stored balances and quotation history before and after the second request.

Expected Result: The server returns an HTTP 409 conflict response because the operation ID has already been used with different allocation details. The original quotation remains unchanged, no additional credits are allocated, and no duplicate history record is created.

Transitions Covered: ST21

TC15 – Quotation History Storage Failure

Precondition: An isolated test environment is available where the Redis quotation history key can be configured with an incompatible data type.

Steps:

1. Configure the selected brand's quotation history key with an incompatible Redis data type.
2. Record the existing credit balances and operation records.
3. Submit a valid quotation request.
4. Examine the server response.
5. Check the credit balances, operation records, and quotation history after the failed request.

Expected Result: The server rejects the request with an HTTP 503 response indicating that quotation history storage is unavailable. No credits are allocated, and no new operation or quotation record is saved. The failure does not cause partial updates.

Transitions Covered: ST22

TC16 – Credit Balance Overflow Prevention

Precondition: An isolated test environment contains an outlet whose existing credit balance is near the supported storage limit.

Steps:

1. Set the outlet's stored credit balance near the maximum supported value.
2. Record the existing balances and quotation history.
3. Submit a quotation that would cause the resulting balance to exceed the supported limit.
4. Examine the server response.
5. Verify whether any balances or quotation records have changed.

Expected Result: The server rejects the allocation with an HTTP 422 response indicating that the resulting credit balance is too large. All outlet balances remain unchanged, and no quotation is saved. The entire allocation is rejected rather than partially processed.

Transitions Covered: ST23

Credit Balance Refresh and Email Delivery

TC17 – Successful Balance Refresh and Email Sending

Precondition: A valid quotation can be generated, the credit balance service is available, and email delivery is configured.

Steps:

1. Enter valid SMS credit amounts and generate a quotation.
2. Confirm that the quotation has been created successfully.
3. Observe the credit balance refresh and email sending operations.
4. Verify that the refreshed balances include the newly allocated credits.
5. Check the email status after sending completes.
6. Confirm that both operations can complete independently.

Expected Result: The updated credit balances are displayed correctly, and the email status changes to Sent after successful sending. Both operations run independently, and neither causes the confirmed quotation or credit allocation to be processed again.

Transitions Covered: ST24, ST25, ST27

TC18 – Credit Balance Refresh Failure

Precondition: A quotation has been generated successfully, but the credit balance service can be made unavailable during the subsequent refresh.

Steps:

1. Generate a valid quotation.
2. Simulate a failure when the feature requests updated credit balances.
3. Observe the displayed balances and error message.
4. Check the generated quotation and stored credit allocation.

Expected Result: The feature displays a message indicating that current balances could not be refreshed. Unavailable balances are shown as — instead of outdated values. The quotation remains valid, and the stored credit allocation is unaffected.

Transitions Covered: ST26

TC19 – Email Failure and Retry

Precondition: A quotation has been generated successfully, and email delivery can be interrupted or configured to fail.

Steps:

1. Generate a valid quotation.
2. Simulate an email sending failure.
3. Check the displayed email status and available actions.
4. Record the stored credit balances and quotation history.
5. Restore the email service and select Retry Email.
6. Check the email status, credit balances, and quotation history after the retry.

Expected Result: The initial failure displays an error and makes Retry Email available. Retrying sends the existing quotation without generating another quotation or allocating additional credits. After successful sending, the email status changes to Sent. Credit balances and quotation history remain unchanged by the email retry.

Transitions Covered: ST28, ST29, ST30

TC20 – Returning to SMS Top Up After Generation

Precondition: A quotation has been successfully generated and its credit allocation has been confirmed.

Steps:

1. Generate a quotation with valid credit amounts.
2. Record the allocated amounts and updated credit balances.
3. Select Back to SMS Top Up.
4. Inspect the credit entry fields and total.
5. Check that the previously allocated credits remain recorded.

Expected Result: The SMS Top Up screen opens with a fresh draft. All credit entry fields are reset, and the total returns to zero. Previously allocated credits remain in persistent storage, and the completed quotation remains available in history.

Transitions Covered: ST31

Quotation History and Persistence

TC21 – Accessing Saved Quotation History

Precondition: A brand has at least one previously generated quotation stored in Redis.

Steps:

1. Open the SMS Top Up screen for the selected brand.
2. Select History.
3. Allow the quotation history request to complete.
4. Check the displayed quotations against the records stored in Redis.
5. Verify that the displayed records belong to the selected brand.

Expected Result: The history screen loads the saved quotations for the selected brand. Each record displays the corresponding quotation information, and quotations belonging to other brands are not included.

Transitions Covered: ST32, ST33

TC22 – Empty Quotation History

Precondition: A brand exists with no previously generated quotations.

Steps:

1. Select the brand without quotation records.
2. Open the SMS Top Up screen.
3. Select History.
4. Observe the history screen after loading completes.

Expected Result: The feature displays a message indicating that no quotations are available. The history screen remains accessible, and no non-existent or unrelated quotations are displayed.

Transitions Covered: ST34

TC23 – History Loading Failure and Recovery

Precondition: No quotations are available locally for the selected brand, and the quotation history service can be made unavailable.

Steps:

1. Make the history service unavailable.
2. Open History for the selected brand.
3. Observe the error message and available actions.
4. Restore the history service.
5. Select Retry History.
6. Check the records displayed after the request succeeds.

Expected Result: The feature displays an error message with a Retry History option when the request fails. After the service is restored, retrying retrieves and displays the saved quotation records without affecting existing balances or allocations.

Transitions Covered: ST35, ST37

TC24 – History Failure with Locally Available Quotations

Precondition: At least one quotation has been successfully generated during the current session, and the quotation history service can be made unavailable.

Steps:

1. Generate a valid quotation and confirm its successful allocation.
2. Return to the SMS Top Up screen.
3. Make the quotation history service unavailable.
4. Select History.
5. Inspect the records and messages displayed.

Expected Result: The feature displays the quotations available from the current session despite the history request failing. A notice explains that the complete saved history could not be retrieved. No additional credits are allocated or quotation records created.

Transitions Covered: ST36

TC25 – Viewing Saved Quotations and Navigation

Precondition: A brand has at least one saved quotation in its history.

Steps:

1. Open History for the selected brand.
2. Select a previously generated quotation.
3. Examine the quotation details.
4. Attempt to modify the displayed credit allocations.
5. Select Back to History.
6. Select Back to SMS Top Up.

Expected Result: The selected quotation opens in read-only mode and displays its original allocation details. The user cannot modify the saved quotation. Returning to history displays the quotation list, while Back to SMS Top Up returns to the credit entry screen. These actions do not change stored credit balances or quotation records.

Transitions Covered: ST38, ST39, ST40

TC26 – Credit Balance and Quotation History Persistence

Precondition: A brand has saved quotations and existing credit balances, with at least one successfully completed allocation.

Steps:

1. Record the brand's existing credit balances and saved quotation history.
2. Enter new credit amounts without generating a quotation.
3. Refresh the browser page.
4. Select the same brand and return to the SMS Top Up screen.
5. Check the credit balances and unfinished credit entries.
6. Open History and compare the saved quotations with the records before the refresh.

Expected Result: The previously allocated credit balances remain intact after refreshing the application. Saved quotations are retrieved from persistent storage without missing or duplicate records. The unfinished credit entries are cleared because drafts are retained only within the current application session.

Transitions Covered: ST41

Error and Recovery Testing

Error and recovery testing examines whether the SMS Credit Reload feature can handle failures without losing confirmed transactions, allocating duplicate credits, or displaying misleading information. The relevant failure scenarios are covered in the test cases above, while this section focuses on how the feature protects data and recovers from those failures.

Validation and Request Failures

Invalid credit amounts and rejected requests should not result in changes to stored balances or quotation history. Users must be able to correct invalid entries and submit a new request once the issue is resolved.

Network failures and server errors during quotation generation require additional safeguards because the server may have completed an allocation before the response was lost. In these situations, the feature retains the original operation ID and prevents changes to the pending request. Retrying the same operation allows the result to be confirmed without allocating the credits again.

Data Integrity

Credit allocation and quotation storage must remain consistent. If quotation history storage is unavailable or an allocation would exceed the supported balance limit, the entire operation should be rejected before any changes are committed.

Requests using an existing operation ID with different allocation details must also be rejected. This prevents conflicting requests from modifying a previously processed transaction.

Independent Operations and Recovery

Once a quotation has been successfully generated, failures in credit balance refreshing or email delivery must not reverse the confirmed allocation.

If balance refreshing fails, unavailable values should be displayed rather than outdated balances. If email delivery fails, the existing quotation can be sent again without generating a new quotation or allocating additional credits.

Quotation history failures should provide a retry option. Where quotations from the current session are available, they may be displayed temporarily, with a clear indication that the complete saved history could not be retrieved.

Transition Coverage Matrix

The matrix links the identified state transitions to their corresponding test cases. This helps check that each transition has been addressed without repeating the same test unnecessarily.

Feature Area	State Transitions	Test Cases
Initial Loading and Brand Selection	ST01–ST07	TC01–TC04
SMS Credit Entry and Validation	ST08–ST14	TC05–TC09
Quotation Generation and Credit Allocation	ST15–ST23	TC10–TC16
Credit Balance Refresh and Email Delivery	ST24–ST31	TC17–TC20
Quotation History and Persistence	ST32–ST41	TC21–TC26

All 41 identified transitions are mapped to the 26 test cases. Related transitions are combined where they can be checked within the same sequence, such as a failed request followed by a successful retry.

This matrix represents the **planned test coverage**. It does not indicate that the test cases have been executed or passed. Actual results would need to be recorded separately during testing.